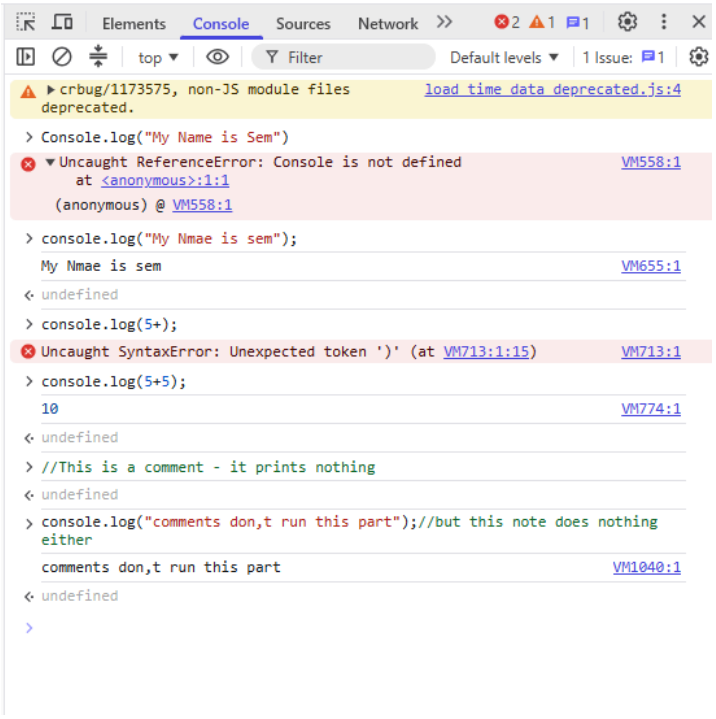
Part 1



Predicted Output

1. My Name is Sem
2. 10
3. It prints nothing
4. Comment don,t run this part

Task - 1.1



If i remove the closing quote, JavaScript will produce a **SyntaxError** because the string is not properly closed. After reading the error, I add the missing quote and run the line again.

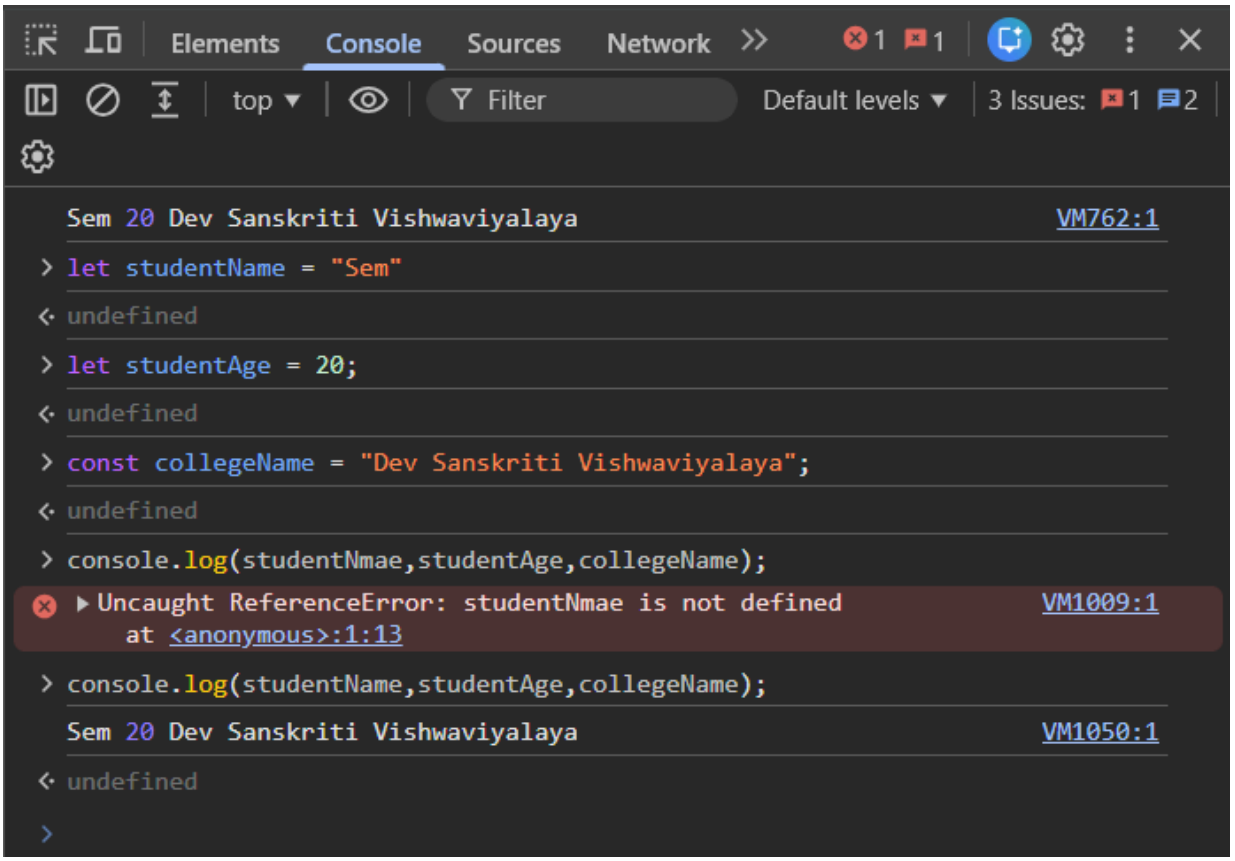
Task - 1.2

```
> let test = "Hello";
< undefined
> console.log(test);
  Hello                                                    VM1363:1
< undefined
> console.log(Test);
✖ ▶ Uncaught ReferenceError: Test is not defined           VM1473:1
      at <anonymous>:1:13
```

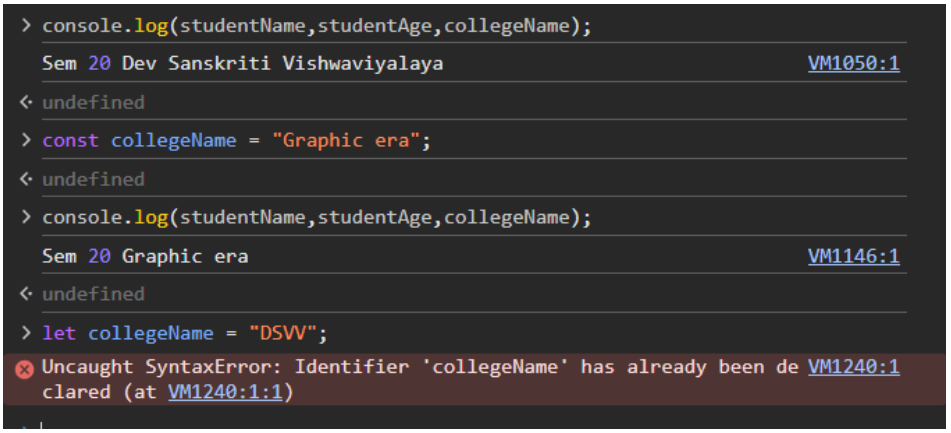
JavaScript is case-sensitive, so test and Test are treated as different names.

Part 2

Task - 2.1

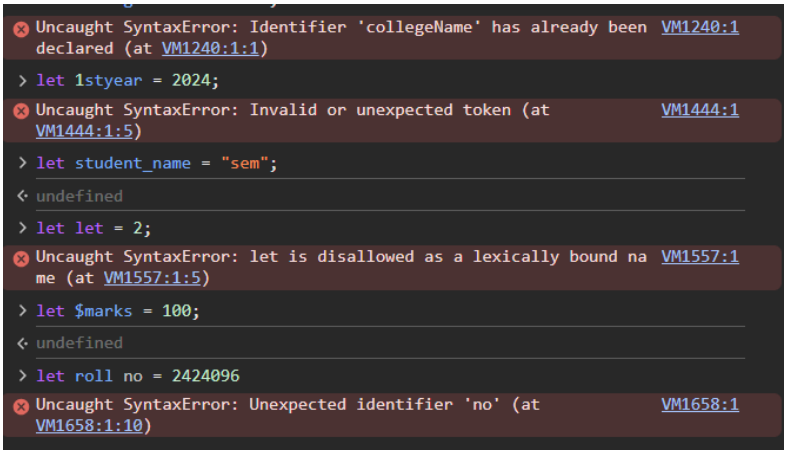


Task - 2.2



When I will reassign the collegename with different value. It will show error - **SyntaxError: Identifier ‘collegeName’ has already been declared**

Task - 2.3



Decleration	Result	Why?
let 1stYear;	Invalid	An Identifier cannot start with a number.
let Student_name	valid	Underscore is allowed on variable name .
let let:	Invalid	‘Let’ is reserved JS keyword it cn=an bo not used as variable name.
let $mark	valid	$ is allowed at the beginning of identifier.
let roll no;	Invalid	Space is not allowed between variable name.

Task - 2.4

```
undefined
> if (true) {
      var a = "i am var";
      let b = "i am let";
  }
undefined
> console.log(a);
  i am var                                                    VM2007:1
undefined
> console.log(b);
  ▶Uncaught ReferenceError: b is not defined                  VM2060:1
      at <anonymous>:1:13
>
```

Because **var** & **let** have different type of scope in JavaScript :-

var - it is global scoped, it is accessible outside the block.

let - it is block scoped, it exist only in side the block.

Part - 3

Task - 3.1

Observation :

I found that when i run

let previousmayor = null;

console .log(typeof previousMayor);

i found it will show object but it is null data type and i read about this. it is a historical bug in JavaScript

```
> let name  = "sem";
undefined
> console.log(typeof name);
  string                                                      VM2254:1
undefined
> let age = 20;
undefined
> console.log(typeof age);
  number                                                      VM2368:1
undefined
> let isStudent = true;
undefined
> console.log(typeof isStudent);
  boolean                                                     VM2387:1
undefined
> let address;
undefined
> console.log(typeof address);
  undefined                                                   VM2402:1
undefined
> let bigNumber = 1234567890123456789on;
undefined
> console.log(typeof bigNumber);
  bigint                                                      VM2424:1
undefined
> let previousMayor = null;
undefined
> console.log(typeof previousMayor);
  object                                                      VM2471:1
```

Task - 3.2

```
> const student = {
  name: "Rohit",
  rollNo: 21,
  course: "BCA"
  };
< undefined
> console.log(student.name);
  Rohit                                                    VM3147:1
< undefined
> const subject = ["javaScript", "DBMS", "Networking"];
< undefined
> console.log(subject[0], subject[2]);
  javaScript Networking                                    VM3391:1
```

Add an isPassing property to the student object and print it.

```
> student.ispassing = true;
< true
> console.log(student.ispassing);
  true                                                     VM3604:1
```

add a 4th subject to the array and print it.

```
> subject.push = ("Operating System");
< 'Operating System'
> console.log(subject[3]);
  undefined                                                VM3797:1
< undefined
> subject.push("Operating System");
✖ ▶ Uncaught TypeError: subject.push is not a function     VM3877:1
      at <anonymous>:1:9
> const subjects = ["JavaScript", "DBMS", "Networking"];
< undefined
> subjects.push("Operating System");
< 4
> console.log(subjects[3]);
  Operating System                                         VM3889:1
```

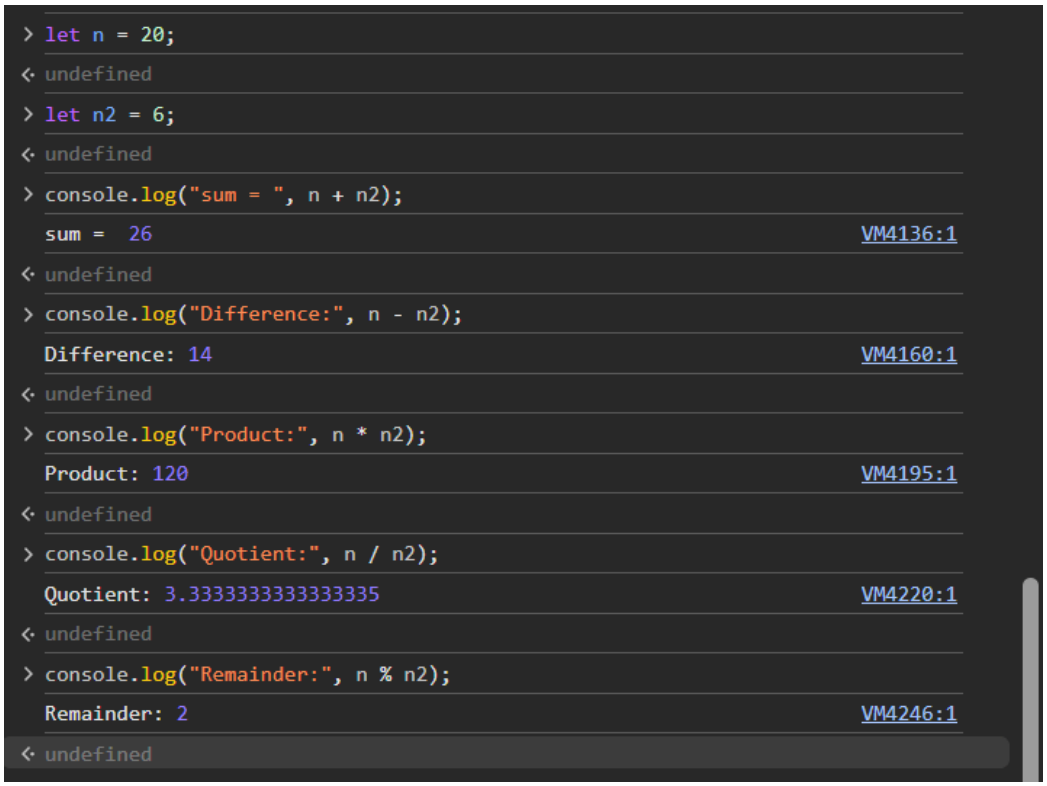
Task - 3.3

```
  bigint                                                   VM2424:1
< undefined
> let previousMayor = null;
< undefined
> console.log(typeof previousMayor);
  object                                                   VM2471:1
```

i found that it will show object but it is null data type and i read about this. it is a historical bug in JavaScript.

Part - 4

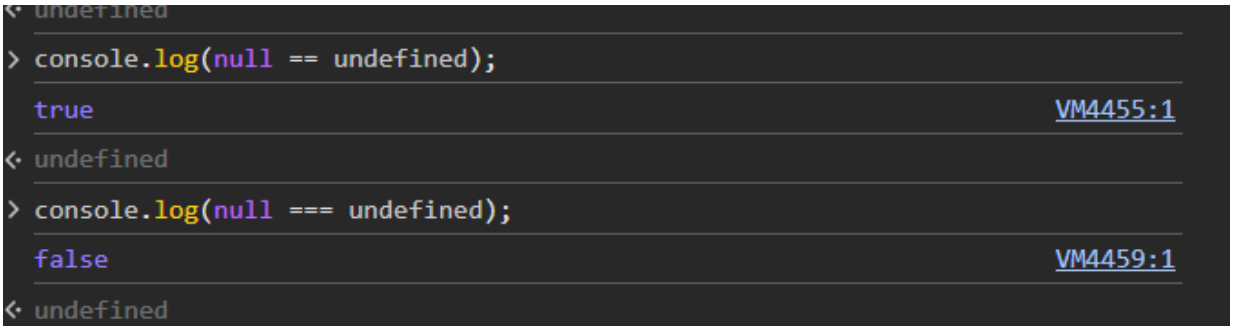
Task - 4.1

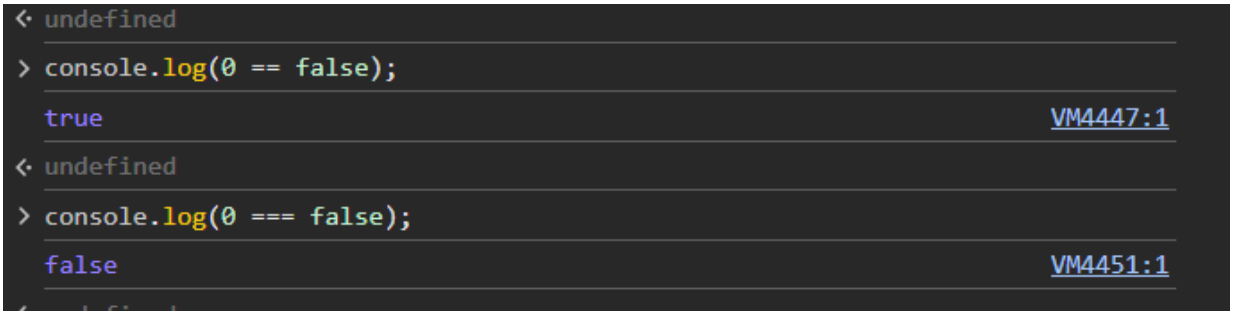


Task - 4.2



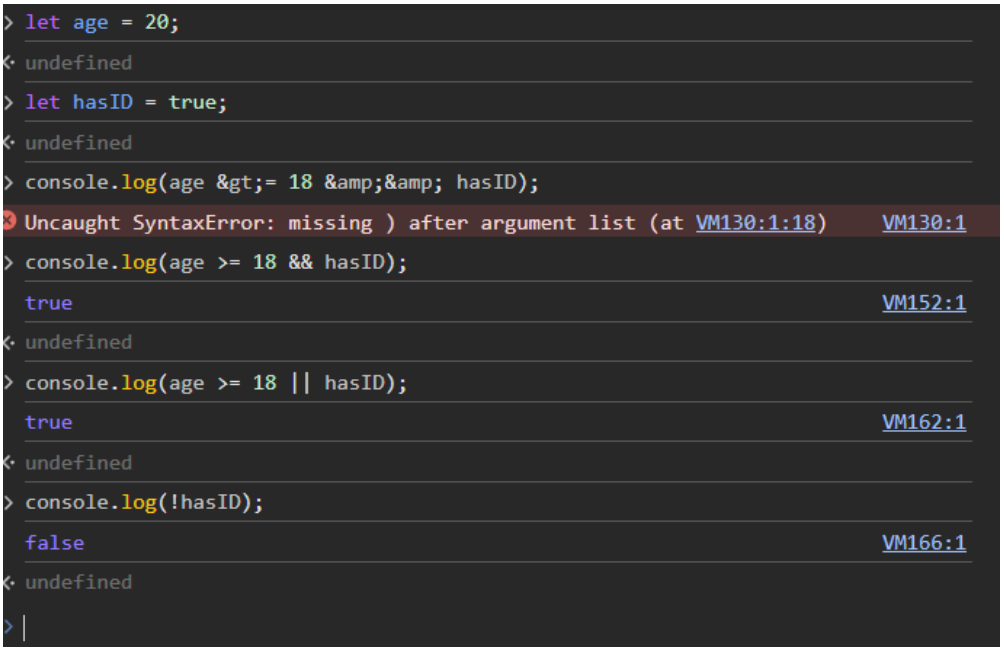
Expression	Result	Why?
5 == "5"	true	== performs type conversion
5 === "5"	false	=== checks both value and data type so it is false 5 is Integer & “5” is string





Expression	Result	Why?
0 == false	true	converts `false` to `0`, so `0 == 0` is true.
0 === false	false	`because` 0 is Integer, while false is boolean
null == undefined	true	javaScript's == has a special rule that considers `null` and `undefined` equal
null === undefined	false	Because they are different type

Task - 4.2



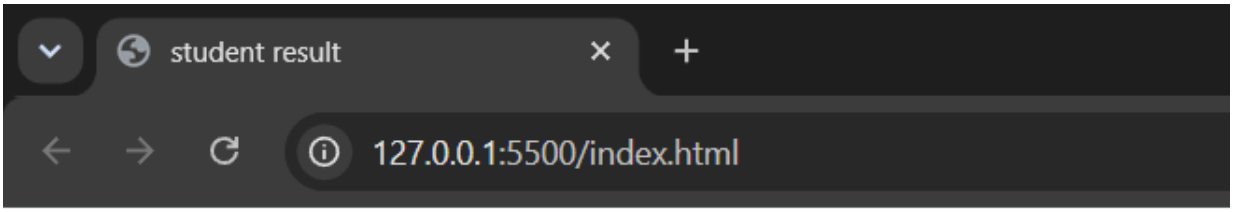
```
> let age = 20;
< undefined
> let hasID = false;
< undefined
> console.log(age>= 18 && hasID);
  false                                                    VM467:1
< undefined
> console.log(age >= 18 || hasID);
  true                                                     VM483:1
< undefined
> console.log(!hasID);
  true                                                     VM527:1
< undefined
```

I can change the value of the hasID to false therefore first line is false left two are true because && this operator means this need both condition true, and || or operate needs only one condition true, and ! NOT operator means opposite of has ID is True

Task - 4.4

```
> let mark = 55;
< undefined
> console.log(mark >= 40 ? "pass" : "fail")
  pass                                                     VM722:1
< undefined
> let mark = 30;
< undefined
> console.log(mark >= 40 ? "pass":"fail")
  fail                                                     VM918:1
```

Part - 5

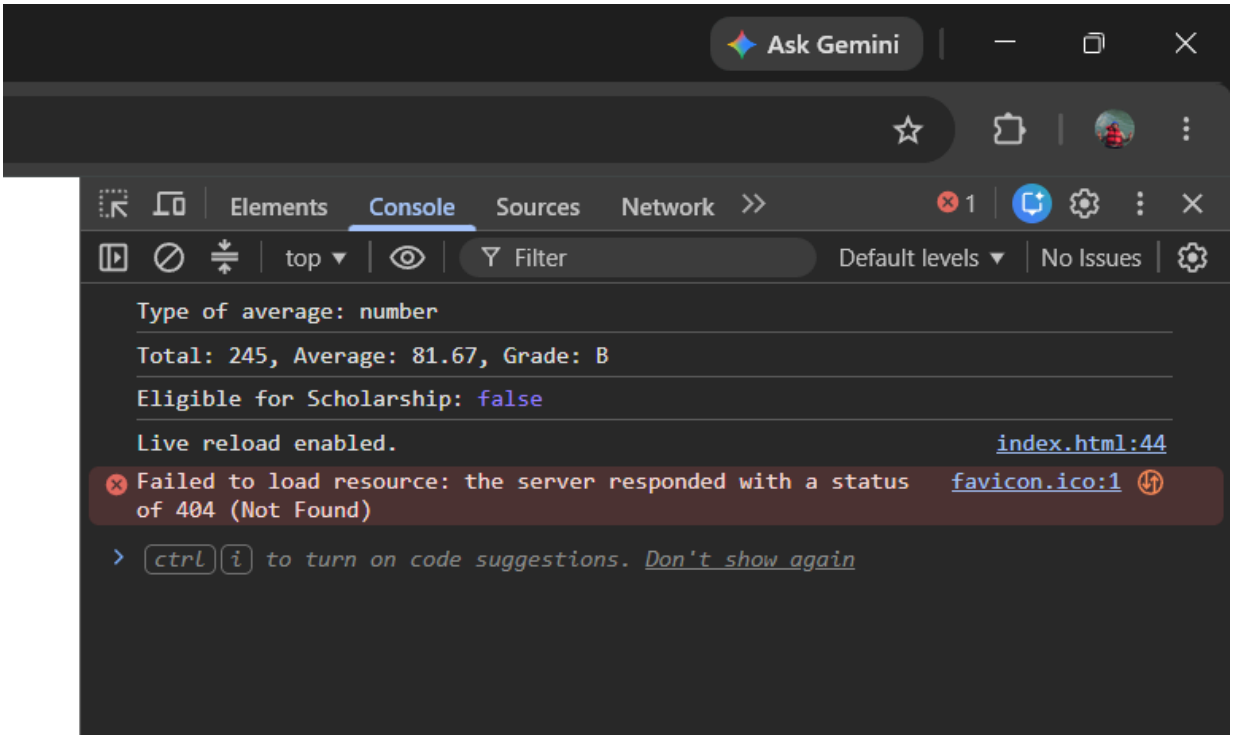


Student Result

Total: 245, Average: 81.67, Grade: B

Average Type: number

Eligible for Scholarship: false



index.html > html

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scal
    <title>student result</title>
</head>
<body>
    <h1>Student Result</h1>

    <p id="result"></p>
    <p id="type"></p>
    <p id="scholarship"></p>

    <script src="script.js"></script>
</body>
</html>
```

Welcome | index.html | script.js

script.js > ...

```javascript
let math = 85;
let science = 78;
let english = 82;

let total = math + science + english;
let average = total / 3;

let grade = average >= 90 ? "A" :
            average >= 75 ? "B" :
            average >= 40 ? "C" : "F";

console.log("Type of average:", typeof average);

let summary = "Total: " + total +
              ", Average: " + average.toFixed(2) +
              ", Grade: " + grade;

console.log(summary);

let attendance = 80;

let isEligibleForScholarship =
    average >= 85 && attendance >= 75;

console.log("Eligible for Scholarship:", isEligibleForScholarship);

document.getElementById("result").textContent = summary;

document.getElementById("type").textContent = "Average Type: " + ty

document.getElementById("scholarship").textContent = "Eligible for
```

Part - 6

Snippet 1

```
> let total = 10 + :
Uncaught SyntaxError: Unexpected token ':' (at VM104:1:18)          VM104:1
> let total = 10 + 5;
< undefined
> console.log(total);
  15                                                                VM254:1
```

Error - **`syntaxError`** — the expression is incomplete because + needs another value.

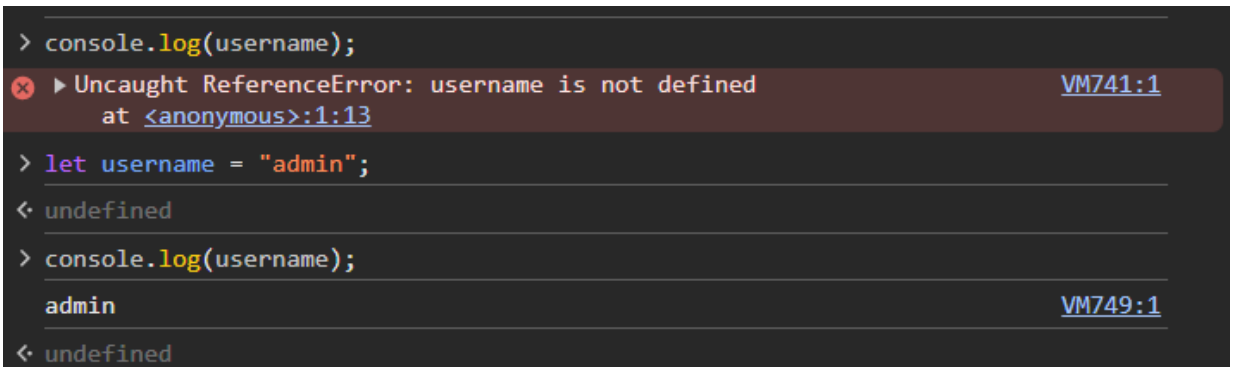
Snippet 2

```
> const rate = 5;
< undefined
> rate = 10;
▶ Uncaught TypeError: Assignment to constant variable.              VM336:1
    at <anonymous>:1:6
```

```
> let rate = 5;
< undefined
> rate 10;
Uncaught SyntaxError: Unexpected number (at VM647:1:6)              VM647:1
> rate = 10;
< 10
> console.log(rate);
  10                                                                VM737:1
< undefined
>
```

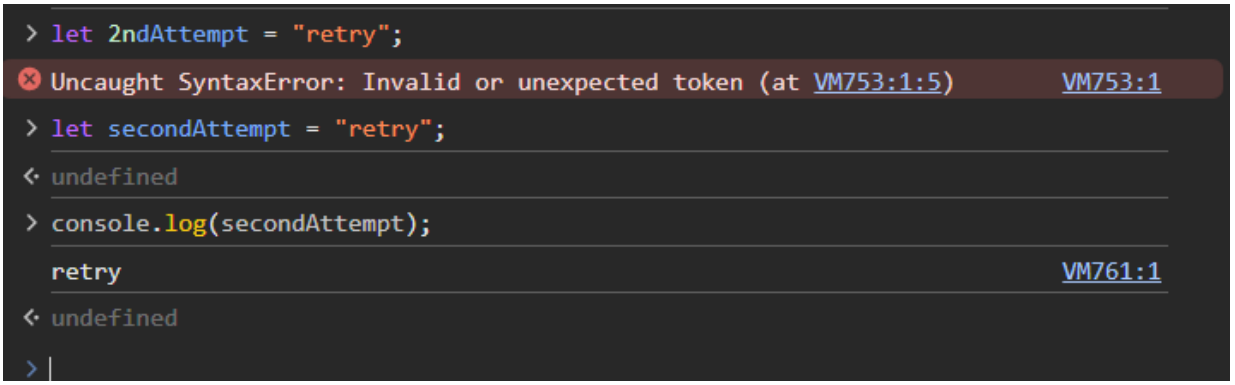
Error - **`TypeError`** — a variable declared with `const` cannot be reassigned.

Snippet 3



Error - `ReferenceError` — `username` is accessed before its `let` declaration is initialized.

Snippet 4



Error - `SyntaxError` — a variable name cannot start with a number

The most confusing part of this lab is part - 5, mini project, simple grade calculator in this i will confused in this code `document.getElementById("result").textContent = summary;` how it works and how we get output it became clear when i search about this how it's work.